

# Keevan Store

## Full Audit, Bug-Fix & Verification Report

Version 2.0 — Post-Fix Verification

| Item                 | Result                                           |
|----------------------|--------------------------------------------------|
| Build Status         | PASS — Clean compilation, zero TypeScript errors |
| ESLint               | PASS — Zero warnings, zero errors                |
| Test Suite           | PASS — 321/321 tests passing                     |
| Critical Bugs Fixed  | 3 — All resolved and verified                    |
| Major Bugs Fixed     | 2 — All resolved and verified                    |
| Minor Bugs Fixed     | 3 — All resolved and verified                    |
| WhatsApp Integration | VERIFIED — All support uses +256 768 345 905     |
| CallMeBot Dependency | REMOVED — No third-party bot required            |
| Report Date          | 12 June 2026, 23:46 UTC                          |

# 1. Bugs Found & Fixed

The following is the complete list of bugs discovered during the comprehensive code audit of the Keevan Store application. Each bug has been fixed and verified with both the production build and the test suite. The audit covered all API routes, UI components, database schema, type definitions, notification systems, and cross-cutting concerns such as currency formatting, authentication flows, and payment processing logic.

| # | Severity | Description                                                                                                                                                                                                                                                      | Fix Applied                                                                                                                                                                                                                             | Status |
|---|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | CRITICAL | IPN ticket event_id references product ID instead of events table ID. The tickets table FK constraint requires events.id, but the IPN handler was passing productId, causing a foreign key violation in real Supabase mode.                                      | Added a lookup query to resolve the events table row from the product_id before inserting the ticket record. Now correctly references events.id.                                                                                        | FIXED  |
| 2 | CRITICAL | Donation orders insert null product_id, violating the NOT NULL constraint on orders.product_id in the Supabase schema. This would cause every donation to fail with a database error in production.                                                              | Updated the Supabase schema to allow NULL product_id on orders (since donations legitimately have no product). Changed the index to a partial index for non-null values only.                                                           | FIXED  |
| 3 | CRITICAL | Withdrawal status mismatch: the Supabase schema defines statuses as 'pending/approved/rejected/paid' but the application code used 'processing' and 'completed'. This would cause CHECK constraint violations on every withdrawal insert or update in real mode. | Aligned the WithdrawalStatus enum, WITHDRAWAL_STATUS_LABELS, mock data, and all UI components to use the schema-defined values: pending, approved, paid, rejected. Updated the admin page and withdrawals page badge logic accordingly. | FIXED  |
| 4 | MAJOR    | Notifications module depends on CallMeBot third-party API. The user explicitly stated no 'call me bot' is needed — just direct WhatsApp support via the phone number +256 768 345 905.                                                                           | Replaced the CallMeBot API integration with console-based notification logging and WhatsApp direct link generation. Removed the CALLEMBOT_APIKEY from .env. Notifications now log admin messages with follow-up WhatsApp links.         | FIXED  |
| 5 | MAJOR    | Landing page JSON-LD structured data injected via dangerouslySetInnerHTML without using the sanitizeForJsonLd utility that exists in utils.ts. This is an XSS defense-in-depth gap.                                                                              | Added the sanitizeForJsonLd import and applied it to the JSON.stringify output before injection into the script tag.                                                                                                                    | FIXED  |
| 6 | MINOR    | WHATSAPP_DISPLAY constant duplicated across contact/page.tsx and site-footer.tsx instead of using the shared export from whatsapp-support.tsx.                                                                                                                   | Exported WHATSAPP_DISPLAY from whatsapp-support.tsx and imported it in both consuming components, removing the local duplicate declarations.                                                                                            | FIXED  |
| 7 | MINOR    | Unused imports: Eye/EyeOff in donation-widget.tsx and ArrowUpRight in withdrawals/page.tsx. These create unnecessary bundle size and lint noise.                                                                                                                 | Removed the unused import statements from both files.                                                                                                                                                                                   | FIXED  |
| 8 | MINOR    | Mock creator-1 (Sarah Creates) had isAdmin: true, which is incorrect. Only the dedicated admin account (nkevinmegan@gmail.com) should have admin privileges.                                                                                                     | Changed creator-1's isAdmin to false. The admin account creator-admin remains the sole admin in mock data.                                                                                                                              | FIXED  |

## 2. Additional Code Changes

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### 2.1 Order Type Update

The **Order** TypeScript interface was updated to allow `productId: string | null` instead of `productId: string`. This change is necessary because donation orders do not have an associated product. The payment success page was also updated to guard against null `productId` when fetching product details, preventing API 404 errors for donation transactions.

### 2.2 Supabase Schema Update

The **orders** table in the Supabase schema was modified to change `product_id UUID NOT NULL` to `product_id UUID (nullable)`. The associated index was changed to a partial index that only indexes non-null `product_id` values, which is both more efficient and semantically correct. This is a manual action that must be applied to the production database (see Section 4).

### 2.3 Test Suite Updates

Two test files were updated to reflect the corrected withdrawal statuses and admin role assignments. The schema validation test now expects four withdrawal statuses (pending/approved/paid/rejected) instead of five, and the auth security test no longer expects Sarah Creates to be an admin. All 321 tests pass after these updates.

### 2.4 .env Cleanup

The `CALLMEBOT_APIKEY` environment variable and its associated comments were removed from the `.env` file. The WhatsApp notification section was renamed from 'WhatsApp Notifications (CallMeBot)' to 'WhatsApp Support' to reflect the new approach. The `ADMIN_WHATSAPP_NUMBER` variable is retained since it is still used for generating WhatsApp direct links.

## 3. Verification Results

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| Check                 | Result | Details                                                    |
|-----------------------|--------|------------------------------------------------------------|
| Production Build      | PASS   | Next.js 16.2.9 Turbopack build completes with zero errors  |
| TypeScript Type Check | PASS   | Zero type errors across all files                          |
| ESLint                | PASS   | Zero warnings, zero errors                                 |
| Unit Tests            | PASS   | 321/321 tests passing (12 test suites)                     |
| Admin Login           | PASS   | nkevinmegan@gmail.com with Keeva#44 authenticates as admin |
| Withdrawal Flow       | PASS   | Statuses aligned: pending -> approved -> paid              |
| WhatsApp Support      | PASS   | All references use +256 768 345 905 direct WhatsApp link   |

|                      |      |                                                           |
|----------------------|------|-----------------------------------------------------------|
| No CallMeBot         | PASS | All CallMeBot references removed from code and .env       |
| JSON-LD Sanitization | PASS | sanitizeForJsonLd applied to landing page structured data |

## 4. Manual Actions Required

The following actions cannot be performed automatically and require manual execution by the platform administrator before the application is deployed to production. These are database schema changes that must be applied via the Supabase SQL Editor.

### 4.1 Make orders.product\_id Nullable

Run the following SQL in the Supabase SQL Editor to allow donation orders to have a null product\_id. This is a safe, non-breaking change — existing orders all have non-null product\_id values, and the partial index maintains query performance for product-based lookups.

```
ALTER TABLE orders ALTER COLUMN product_id DROP NOT NULL;
-- Replace the existing index with a partial index
DROP INDEX IF EXISTS idx_orders_product_id;
CREATE INDEX idx_orders_product_id ON orders(product_id)
WHERE product_id IS NOT NULL;
```

### 4.2 Verify Withdrawal Status Constraint

Confirm that the withdrawals table status CHECK constraint matches the application code. The expected values are: pending, approved, rejected, paid. If the constraint currently includes 'processing' or 'completed', run the following SQL to update it:

```
ALTER TABLE withdrawals DROP CONSTRAINT IF EXISTS
withdrawals_status_check;
ALTER TABLE withdrawals ADD CONSTRAINT
withdrawals_status_check
CHECK (status IN ('pending', 'approved', 'rejected', 'paid'));
```

### 4.3 Deployment Checklist

Before deploying, ensure the following steps are completed in order: (1) Apply the Supabase schema changes listed above. (2) Remove the CALLMEBOT\_APIKEY variable from any production environment configuration. (3) Deploy the updated application code. (4) Verify the admin login works with nkevinmegan@gmail.com and the password Keeva#44. (5) Test a sample withdrawal flow to confirm the status transitions work correctly (pending to approved to paid). (6) Test the WhatsApp support links on the contact page, download page, and dashboard to confirm they all direct to +256 768 345 905.

## 5. Files Modified

| File Path                                   | Change Type                                                             |
|---------------------------------------------|-------------------------------------------------------------------------|
| src/types/index.ts                          | WithdrawalStatus enum updated; Order.productId made nullable            |
| src/lib/constants.ts                        | WITHDRAWAL_STATUS_LABELS aligned to schema values                       |
| src/lib/mock-data.ts                        | Fixed withdrawal statuses and creator-1 isAdmin flag                    |
| src/lib/notifications.ts                    | Removed CallMeBot; replaced with console logging + WhatsApp links       |
| src/components/shared/whatsapp-support.tsx  | Exported WHATSAPP_DISPLAY constant                                      |
| src/components/shared/site-footer.tsx       | Using shared WHATSAPP_DISPLAY export                                    |
| src/components/store/donation-widget.tsx    | Removed unused Eye/EyeOff imports                                       |
| src/app/api/pesapal/ipn/route.ts            | Fixed ticket event_id to reference events table                         |
| src/app/admin/page.tsx                      | Updated withdrawal status badges; getProductName handles null productId |
| src/app/(dashboard)/withdrawals/page.tsx    | Updated status icons/badges; removed unused import                      |
| src/app/payment/success/page.tsx            | Added null check for productId before fetching product                  |
| src/app/contact/page.tsx                    | Using shared WHATSAPP_DISPLAY export                                    |
| src/app/page.tsx                            | Added sanitizeForJsonLd to JSON-LD structured data                      |
| supabase/schema.sql                         | orders.product_id made nullable; partial index                          |
| .env                                        | Removed CALLMEBOT_APIKEY; renamed WhatsApp section                      |
| src/__tests__/lib/schema-validation.test.ts | Updated withdrawal status assertions                                    |
| src/__tests__/lib/auth-security.test.ts     | Updated Sarah Creates isAdmin expectation to false                      |